

Marketing Email Generator

A multi-agent workflow for creating, revising, and sending small-business marketing emails.

Google ADK + Gemini + Cloud Run

Human-in-the-loop review



The problem

Small businesses need polished marketing emails, but creating them often requires more than writing copy.

- Email layout often requires HTML/CSS or template-editing skill.
- Images, content blocks, brand guidance, and CTAs need to fit together.
- Small-business owners need fast iteration without losing control.



Target users: owners and local marketers who need professional output without becoming email designers.

The solution

A conversational app that turns user instructions into a production-ready email.

User input

The user describes the email purpose, desired styling, and content needs.

Agent-built draft

Agents select approved blocks and assets, then generate copy and styling choices.

Revision loop

The user can request changes in plain English or edit HTML directly.

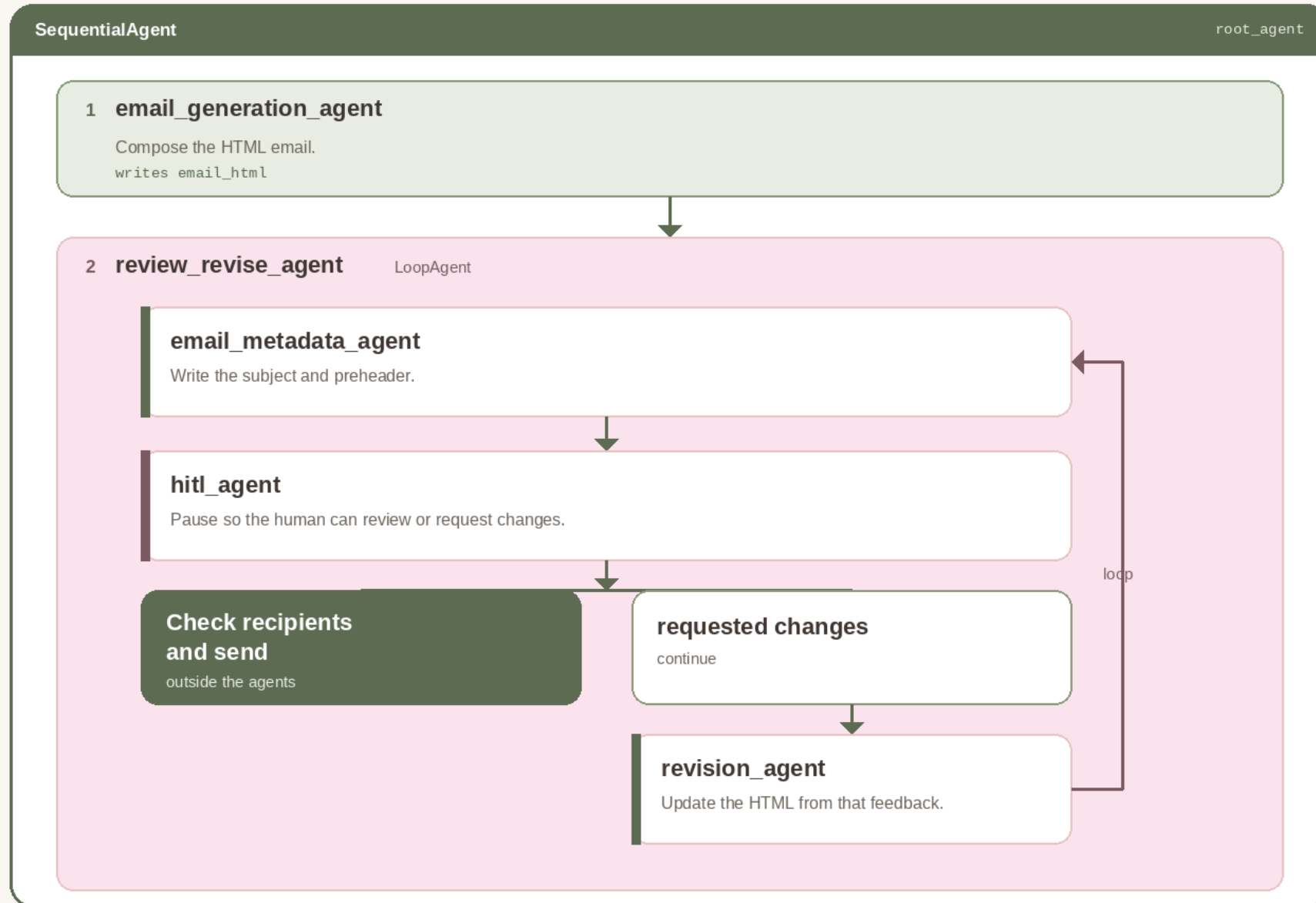
Controlled send

Recipients are checked against unsubscribes before Gmail delivery.



Agent Workflow

agent.py



Controlled creative storehouse

The agents build the first draft from the user's input. Later revisions also use the user's input, but update the same email instead of starting over.

Reusable HTML blocks

Approved layouts for headers, heroes, content, products, CTAs, and footers.

Brand guidance

Colors, font options, spacing choices, button styling, visual direction, and copy guidance.

Product & campaign profiles

Recommended blocks and composition guidance for the requested email.

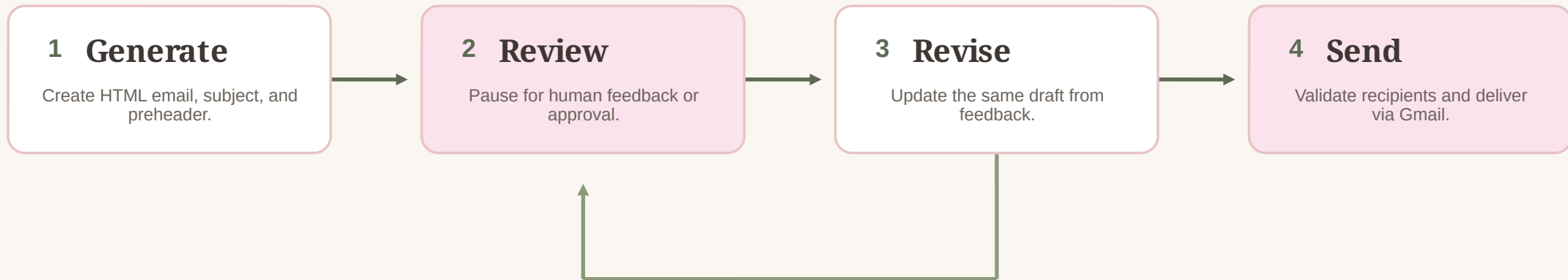
Approved assets

Product images and logo files available to the agents.

The agents use these controlled resources together with the user's instructions to generate the email's copy, styling, layout, and imagery.

Product features

The core product flow is a review loop: generate a draft, review it, revise the same email, and continue until the user is satisfied.

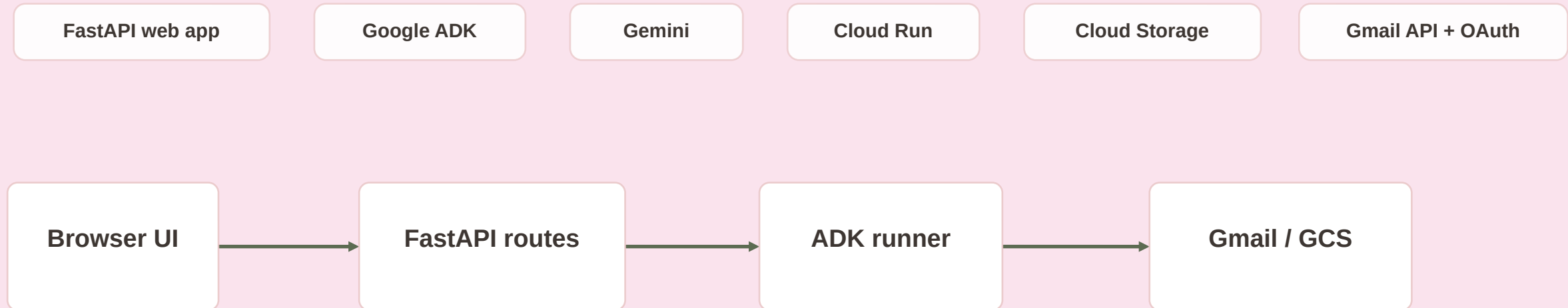


Loop continues only while the user requests changes. Sending happens after the email is satisfactory.

Visible controls: direct HTML edits, subject/preheader changes, recipient validation, Gmail connection, and unsubscribe handling.

Technology architecture

A deployed web application wrapped around a resumable ADK agent workflow.



Cloud Storage persists storehouse assets, drafts, and suppression data. Gmail sending uses user-authorized OAuth.

Impact and roadmap

Impact

- Makes polished email creation accessible to users who do not know HTML or CSS.
- Keeps the human in charge of judgment, revision requests, recipient lists, and final sending.
- Uses approved assets and profiles so output stays closer to the business brand.

Roadmap

- Combine the functionality of this app with another multi-agent workflow I built that designs marketing journeys.
- Extend the project to handle any use case, not just my wife's local business.
- Add persistent session storage for more robust production usage.